SALTDRAG · DESIGN DECISION RECORD

SaltDrag design for construction

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DOCUMENT CONTROL

Document ID	SDG-DDR-002	Version	0.1
Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/saltdrag		
Source	docs/decisions/0002-design-for-construction.md @ c7f0938		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	Changes that make the concept physically buildable, with the reason for each; made under Amish's 2026-10-03 pre-approval

0002: Design for construction

- **Date:** 2026-10-03
- **Status:** accepted. Decided by Amish under his pre-approval of 2026-10-03: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost." The standing instruction for this step (Amish, 2026-09-30) is to "fix the design assumptions to match and be physically feasible".

Context

The concept described portable geared drums "staked on the bunds", a rake "in bolted sections", ropes, bridles and a windrow board, without saying how any part is made or how it joins its neighbours. The constructability review went component by component through how each is made and fixed, and checked the parametric model for overlaps, floating parts and clearances with build123d (`python cad/src/model.py --check` : 63 of 63 checks pass).

Changes

Table 1. Changes from the concept to the constructable design.

#	Component	The concept had	The constructable design has	Why
1	Capstan frame	A portable geared drum, no frame described	Two side frames (6 mm cheek welded on a 40 mm box runner) bolted together by three cross tubes with end plates	Each side frame is under 21 kg and the frame comes apart for carrying (R6); bolted tubes let the drum go in between the cheeks
2	Capstan position	Staked on the bund	On dry ground behind the bund, drum 1.5 m behind the crest centre, with a roller saddle on the crest	The crest is too narrow to stand on while cranking; the roller keeps the rope off the bund
3	Drum	A drum	Welded drum: 168 mm pipe barrel, 330 mm flanges, a brake rim on the left flange, a 40 mm shaft welded through; rope wound over the top	The flanges hold 58 m of rope in four layers; rope leaving the top is level with the roller, so it does not lift the capstan
4	Shaft supports	Not shown	Four-bolt flange bearings bolted to the cheeks, shafts through clearance holes	Bought bearings in stainless and thermoplastic for brine; holes leave 5 to 10 mm of clearance
5	Gearing	"Geared"	Bought module 4 gear (84 teeth, webbed) and pinion (14 teeth), keyed on; the pinion overhangs outside the right bearing	One stage gives 6 to 1; a webbed gear is 10 kg instead of 22 kg solid
6	Ratchet and pawl	A ratchet pawl	20-tooth ratchet wheel on the drum shaft; the pawl on a 40 mm boss welded to the left cheek, with an M24 pivot bolt	A plain pin cantilevered from the cheek would bend at 6 kN (SDG-CAL-001, Section C)
7	Brake	A brake	Differential band brake: a lined steel band round the brake rim, both ends to a lever pivoted through the left cheek	Holds the rated pull with a 92 N hand force; no extra drum needed
8	Cranks	Crank handles	Two removable cranks on hubs with cross pins, set 180 degrees apart, on a 25 mm crank shaft in its own bearings	Removable so they are off whenever rope is paid out under load (SDG-DDR-001, D9)
9	Gear guard	None	A yellow 2 mm steel cup over the gear mesh on three spacers	Covers the nip between gear and pinion
10	Anchoring	"Screw or plate anchors"	An ear on each cheek at rope height, a shackle, chain and turnbuckle to a screw-in helical anchor 1.7 m behind	Chain at rope height keeps the capstan from tipping (1.55 times the tipping moment)
11	Rake beam	Bolted sections	One 3.0 m FRP box beam, 100 x 100 x 6 mm	At 12.8 kg one piece is carried by two people; no splice to wear in brine
12	Blade fixing	Not shown	Ten countersunk M10 stainless bolts through blade and beam, with stainless crush sleeves inside the tube	The sleeves stop the bolts crushing the FRP
13	Skids	Height-setting skids	Three UHMW-PE skids on L-shaped stainless hangers bolted to the beam's back face through slots	The slots set the blade 5 to 25 mm above the floor
14	Bridle fixing	"Bridle and swivels"		

#	Component	The concept had	The constructable design has	Why
			Stainless end caps on the beam ends with welded eye plates; bridles shackled to the eyes, front and rear	Pulls the beam at its ends through a part made for the load; the beam's bending stays at 28 MPa at 6 kN
15	Ballast	None	Two woven bags of about 24 kg of salt strapped on the beam behind the blade	Without them the rising rope lifts the blade 5.9 m from the bund toe; with them, 3.3 m (SDG-CAL-001, Section D)
16	Windrow board	A board at the bund edge collects the salt	Removed	The rake lifts before it reaches the bund edge, so a board there would collect nothing; the windrow forms about 3.3 m from the toe
17	Rope dampers	None	A weighted bag on each loaded rope	Snap-back protection (SDG-DDR-001, D10)

What did not change

What the product does, its pitch and its safety case are unchanged: two hand capstans on the bund side wind a rake across the pan and nobody enters the brine. The windrow position (change 16) and the ballast (change 15) follow from physics the concept had not checked, and both were made under the pre-approval.

Consequences

- New BOM lines: 6b (25 mm bearings), 11b (chains and turnbuckles), 12 (bund roller saddle), 20 (rope dampers), 22 (ballast bags), 23 (fabrication labour and galvanising).
- General arrangement SDG-DWG-001 goes to Rev P2; making sketches SDG-DWG-101 to 113 are new.
- Concept media, STEP and STL files are regenerated from the constructable model.
- `design_state: constructable` in `project.yaml`.